

Governance Memo

To: Hypothetical Court Auditor

Subject: Explainability Review of COMPAS Replacement Model

This memo summarizes what the explanation methods reveal about the COMPAS replacement model and what governance actions are warranted. The model appears to rely most heavily on factors such as prior arrests, age category, charge degree, and other encoded predictors associated with recidivism risk. Across the reviewed individuals, SHAP and LIME often agreed on the main drivers of a prediction, which increases confidence that those features were materially influencing the local decision. However, the methods did not always align perfectly. In some cases, they differed in feature ranking, contribution size, or which correlated variable appeared most influential. This suggests that the model's local behavior can be sensitive to the explanation method and to the structure of the surrounding feature space.

These findings should be interpreted with caution. LIME provides a local approximation rather than a full description of model behavior, while SHAP offers a more systematic attribution framework but can still be affected by correlated features and approximation choices. Counterfactual explanations are useful for identifying what changes could reverse a prediction, but some counterfactuals may be implausible or may rely on features that should not be considered actionable in practice. For that reason, explanation outputs should be treated as diagnostic tools, not as conclusive proof that the model is fair, lawful, or reliable.

I recommend continued monitoring in three areas. First, the court should require regular subgroup performance audits, especially false positive and false negative rates by race and sex. Second, explanation monitoring should be repeated on a scheduled basis to detect drift in the features driving decisions. Third, any counterfactuals that require immutable or protected attributes should be flagged and excluded from recourse guidance. Together, these controls would support a more defensible and transparent deployment posture.

Approx. word count: 287